STRUCTURE & FUNCTIONS OF CYBER-PHYSICAL SYSTEMS

Design Thinking Field Manual

Memorize the whole deck the easy way: one beach story, the exact exam points, and a memory hook on every topic.

Module **MMC-01a** Chapter **5 · Ideation** Lecturer **Prof. Dr. M. Górka**

Source **5.pptx · 48 slides**

💡 Meet Maya — your memory anchor

Maya is an inventor on the beach who wants to build a gadget for **Jack, a 54-year-old Baltic-Sea lifeguard** who loves surfing and hates littering tourists. She has **no idea what to build yet** — and that's exactly the point.

Every single slide just answers one question: **“How does a smart inventor go from a fuzzy problem to a gadget the customer actually loves — by failing fast and cheap?”** Keep Maya chasing Jack down the beach and the chapter tells itself.

★ MASTER KEY → Walk from the 🏖️ Problem beach into the 🌊 Solution sea: Understand · Observe · Design-principles · Ideate · Prototype · Test

00 · THE SHAPE OF THE WHOLE CHAPTER

See the skeleton before the bones

Design Thinking is **step 5 (Ideation) of System Engineering:**

```
1 Basics → 2 Requirements → 3 Risk → 4 Change → 5 IDEATION ★ → (6 Design) →
(7 Test) → (8 Project Setup)
```

The method is a **6-step loop** that splits cleanly in half — first you understand the **problem** (on dry sand), then you build & test a **solution** (in the water). No acronym needed: just watch Maya walk from the beach into the sea.

> 🧠 🏖️ **Problem space → 🌊 Solution space.**
> Understand → Observe → Design principles → Ideation → Prototyping → Testing.
> (Persona = the bridge from sand to sea.)

THE TRICK · WATCH, DON'T MEMORIZE

🎬 The Memory Movie

Don't memorize sentences — your brain hates random letters. Instead **close your eyes and play this silly 6-scene movie** of *Maya builds Jack's beach gadget*. Each picture *is* the point, so when you replay the movie, the exam answer just appears. Read it 3 times, then try to replay it with your eyes shut.

SCENE 1 · 🏖️ PROBLEM SPACE

Maya stands on the sand holding a **giant jigsaw piece with a "?"** — grasping the puzzle: what's the problem, what's the goal (NOT the gadget yet!). → (1) **UNDERSTAND.**

SCENE 2 · 🏖️ PROBLEM SPACE

She lifts a **telescope and spies on Jack** rescuing swimmers, asking about his hobbies & family. → (2) **OBSERVE the customer.**

SCENE 3 · THE BRIDGE

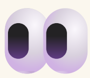
She glues her notes onto a **life-size cardboard "Fake-Jack"** she can interrogate. → (3) **DESIGN PRINCIPLES = build a Persona.**

SCENE 4 · 🌊 SOLUTION SPACE

A **storm of crazy lightbulbs** rains down — the wilder the better, no rules. → (4) **IDEATION.**

SCENE 5 · 🌊 SOLUTION SPACE

She tapes **cardboard & duct-tape** into a rough gadget and hands it over **without a word.** → (5) **PROTOTYPING (simpler = better).**



SCENE 6 · 🌊 SOLUTION SPACE

She **hides behind a palm tree and just watches** Jack fumble with it, taking notes. → (6) **TESTING — then loop back to Scene 5.**

🎥 Replay in one breath: **puzzle piece → telescope → cardboard Fake-Jack → lightbulb storm → duct-tape gadget → spy from the palm tree.** Scenes 1–2 = Problem beach, 4–6 = Solution sea, Scene 3 is the bridge.

🎯 The 3 things design thinking keeps in focus

Picture Maya's surfboard balanced on three rocks — take one away and it tips over:

❤️

Customer / benefit

Does it actually help Jack? **Desirability** — the customer is the centre of everything.

🔧

Possibility to realize

Can we actually build it? **Feasibility.**

💶

Commercial viability

Will it sell / pay off? **Viability.**

🖼️ Picture-hooks for the tricky bits

Same trick for everything that wouldn't stick — turn it into one strong, weird image:

🌈

Heterogeneous team

A beach-volleyball team where **everyone looks totally different** (chef, child, granny, robot). **Mixed crowd = better ideas.**

💸

Fail fast & cheap

Maya snaps a **cardboard gadget in half**, laughs, grabs another. **Failure costs almost nothing.** Agile + iterative.

🤫

Observe rules

Maya **zips her own mouth shut** and points the mic at Jack. **Let the customer talk; don't sell your idea.**

📣

Extreme voices

The **loudest, weirdest beach-goer** teaches Maya more than 100 calm tourists. **Extremes reveal the real needs.**

Competency 0→5

A 6-rung **lifeguard tower**: ground = **0** (**nothing**), top platform = 5 (**expert who invents new moves**).

Persona vs Buyer Persona

A Greek **theatre mask** (Persona = a role/face) a **shopkeeper** wears to play “the perfect customer” (Buyer Persona). **Write Jack like a novel character.**

Persona card

One big ID card: **photo, stats, 5 tech apps**, and four arrows — 😊 Motivates, 😠 Frustrates, 🎯 Goals, 🤕 Pain-points.

Decision tools

To pick the best idea, Maya spins a **slot machine of boxes** (**Morphological box**) and a **QFD scorecard**.

The MRI win

A **scary white tube** repainted as a **pirate submarine** so kids climb in laughing. **Solve the real fear, not the machine.**

Ideation rule

In the workshop you **only draw** — **no writing, no sharing**, min. 5 (better 10) wild ideas.

🧠 **How to actually lock it in:** (1) read a scene, (2) **shut your eyes and re-see the picture**, (3) say the real meaning out loud. Do the 6-scene movie + picture-hooks twice today and once tomorrow — then test yourself with the flashcards. Seeing beats reading; testing beats re-reading.

THE FULL LESSON · ALL SLIDES

Taught like a story — with the points you must write

01 What *is* Design Thinking?

STORY

Maya doesn't start by drawing a gadget. She asks, "what would actually help Jack — and would anyone build it or buy it?" Design thinking is a whole **method for inventing creative ideas and solving problems**, and the trick is to be **agile**: try, fail cheaply, try again. Her surfboard reads **"Fail fast and cheap."**

WRITE IN EXAM — DEFINITION

- **Design Thinking = a method for developing creative ideas and for problem solving.**
- Puts **3 things in focus**: the **customer / benefit**, the **possibility to realize** (feasibility), and **commercial viability**.
- The **team** should be **as heterogeneous as possible**.
- It is an **agile method** needing **several iterations** to reach the goal.
- Motto: **➜ Fail fast and cheap!**

Surfboard on 3 rocks = Customer · Feasibility · Viability. Mixed team + fail fast & cheap.

02 The Process — two spaces, six steps

STORY

Picture the beach. On the **dry sand** (**Problem space**) Maya figures out *what's wrong and who for*. In the **water** (**Solution space**) she *makes things and tests them*. She crosses the wet line many times — it's a **loop, not a straight line**.

📝 **WRITE IN EXAM — PROCESS**

- Two spaces: **Problem Space** (understand the problem & customer) and **Solution Space** (create & test solutions).
- Six steps in order: **(1) Understand → (2) Observe → (3) Design Principles → (4) Ideation → (5) Prototyping → (6) Testing**.
- The process is **iterative** — loop through prototyping & testing several times.

🧠 **Walk from the beach into the sea.** Land = Understand + Observe · Water = Ideate + Prototype + Test · Persona = the bridge.

03 (1) UNDERSTAND — grasp the puzzle

🧒 STORY

Maya holds a **jigsaw piece with a big "?"**. First she builds a **team**, then asks: what should we make new? what's the *problem*? what's the **target condition** after Jack uses it — the *outcome*, **NOT the gadget itself!** Who's the customer? What's the environment (wind, salt, sand)?

📝 **WRITE IN EXAM — UNDERSTAND**

- **Content: understand the problem and the target. Build the team.**
- What should be **newly developed**? What's the **problem**?
- What's the **target condition** after using the tech — **NOT the technical solution!**
- **Who is the customer?** Which **environment conditions** matter?
- **How to:** an **initial expert circle / brainstorming**, **updated with each new info** during the process.

🧠 Puzzle piece = describe the **goal/outcome, not the gadget.**

04 (2) OBSERVE — spy on the customer

STORY

Maya grabs a **telescope** and follows Jack. She doesn't pitch ideas — she **shuts up and listens**. She asks about his *real* problem, what it must do, and even his **family & hobbies**. She talks to the **loud**, **extreme beach-goers** because they shout out the useful stuff calm tourists never mention.

WRITE IN EXAM – OBSERVE

- **Content: understand the customer** — their **will**, **wishes & inner needs**; **collect as much info as possible** (actual problem, what it must do, *personal background*).
- **How to: learn from people** — **talk & interview**, **spend time with / observe**, find the relevant **peer groups**; **extreme voices** reveal more than the average customer.
- **Important: let the customer talk, not the interviewer! Don't sell your own idea!**

Zipped mouth + telescope. The customer talks; you only listen. Loudest voice = best info.

04a Competency level of the persona (0–5)

STORY

When Maya rates how much Jack knows about a technology, she climbs a **6-rung lifeguard tower**: ground = **0** (**nothing**), up to the top **5** (**Expert** who invents new state-of-the-art moves).

0 = ground, 5 = top of the lifeguard tower.

05 (3) DESIGN PRINCIPLES — build "Fake-Jack" (the Persona)

STORY

Maya glues all her notes onto a **life-size cardboard cut-out of Jack** — a **fake customer** she can ask questions to. She finds the **common patterns** across the people she met, gives him a photo, job, hobbies, fears, goals, and **updates her "Understand"**. Now when unsure she just asks the cardboard: "Jack, would you like this?"

📝 **WRITE IN EXAM — DESIGN PRINCIPLES**

- **Content: synthesis of steps 1 & 2. Find patterns** → the **common sense** between customers.
- Create a **profile** using **"Personas"** or **"Point-of-View"**; **visualize** the customer's view.
- **Update your "(1) Understand"** (problem, target, functions, features).

📝 **WRITE IN EXAM — PERSONA FACTS**

- **Persona** = concept from **analytical psychology** describing the **outer view/role** of a person (Greek/Latin = a **face/role**).
- **Buyer Persona** = persona **adapted to economics** to **understand the customer**; a **fictional but fitting person** for the target group; describe them **fully (attributes, background, behaviour) like a good actress**; lets you **decide in their name** ("would he use it?"). ➜ **write them like a novel character.**

🧠 **Cardboard Fake-Jack** = a fictional customer you interrogate. **Theatre mask** = persona (a role). Write him **like a novel character.**

06 (4) IDEATION — the lightbulb storm

🧑 STORY

Now the fun part. **Crazy lightbulbs rain from the sky** — Maya wants the **wildest ideas possible**, with **NO limits** on cost or tech. First everyone draws **alone**, then they **gather & discuss**, then they **pick** the best with a **tool** (a slot-machine **morphological box** or a **QFD** scorecard), and finally they **show the customer for feedback.**

📝 **WRITE IN EXAM — IDEATION**

- **Content: create ideas to solve the customer's pain points.**
- **NO boundaries** (cost, tech...) — **the crazier the better!**
- **Consolidate** ideas → **decision making** → **get customer feedback** → if needed, **competitor analysis**.
- **How to: individual** creation → **gather & discuss** → **decide via tools** (**morphological box**, **QFD**, ...) → **customer feedback**.
- Selection criteria: **practicability** · **efficiency** · **economy**.

🧠 **Lightbulb storm = no limits, crazier is better.** Pick with the **morphological box / QFD slot machine.**

07 (5) PROTOTYPING — duct-tape it, say nothing

STORY

Maya tapes together a **rough cardboard gadget** — **the simpler the better** — and hands it to Jack **without saying a single word**. If she has to explain it, it failed. It must **solve Jack's problem**, not show off **her** idea.

WRITE IN EXAM – PROTOTYPING

- **Content: create a tangible prototype** — a **physical prototype that can be experienced.**
- **Everything is allowed; the simpler the better!**
- **Important: NO verbal explanation to the customer!** The **customer must understand the solution, not you.** It must **solve the customer's problem, not realize your idea.**
- **How to:** be creative!

🧠 **Duct-tape gadget + zipped mouth.** Simple, silent, customer-first.

08 (6) TESTING — spy from the palm tree, then loop

STORY

Maya **hides behind a palm tree** and just **watches** Jack fumble with the gadget — **no demo, no help** — taking notes on what to fix. If it flops, she can **change or even throw the idea away** next round. Then she **loops back to prototyping. Take several iterations!**

WRITE IN EXAM – TESTING

- **Content: customer tests the prototype** → target is to **get feedback.**
- If misunderstood / not working → **change or discard** the idea next cycle.
- **Observe the customer** during tests; note improvements. **Take several iterations!**
- **How to: NO demonstration! Visually observe** the customer using it. **Iterate!**

🧠 **Palm-tree spy.** Hand it over, shut up, watch, loop.

09 The MRI example (the exam's favourite story)

🧒 STORY

A real win: an **MRI scanner terrified children**. Instead of fixing the *machine*, designers fixed the *fear* — they redesigned the room as a **pirate-submarine adventure** so kids happily climbed in. That's design thinking: solve the **customer**'s **real problem**, not the technical one.

📝 WRITE IN EXAM

- Successful design thinking example — an **MRI was scary for children, so it was redesigned** to be friendly/playful and accepted by kids.
- Lesson: customer-centred thinking beats a purely technical fix.

🧠 **Scary white tube → painted pirate sub.** Fix the fear, not the magnet.

10 The workshop exercise + Summary

🧒 STORY

In class you *run* the whole loop in **5 teams**. The big rules: interview **other** groups (never your own), in Ideation you **only draw — no words, no sharing**, and you **hand prototypes over silently** and just watch.

📝 WRITE IN EXAM — KEY RULES & SUMMARY

- **Observe:** interview **min. 2 customers from other groups**, not your own; describe the problem briefly only.
- **Ideation-create: drawing only, NO writing, NO exchange**, min. 5 (better 10) ideas; **select** best 5 via a system (morphological box).
- **Prototyping:** pick **2 ideas**, build **physical** prototypes; hand over with no explanation.
- **Summary:** design thinking is a **customer-centred method for designing products**; in **fast, iterative cycles** you **analyse & solve** a customer's needs with several ideas.

🧠 **Draw-don't-write, interview strangers, hand over in silence.** Stay customer-centred · fail fast & cheap · iterate.

YOUR PROGRESS

0 / 10 topics learned

ACTIVE RECALL · FLIP TO TEST YOURSELF

Flashcards

Read the question, say the answer *out loud*, then flip. Cover the deck and run it twice — that's how it sticks.

All flashcards (Q/A)

1. What is Design Thinking?
A method for developing creative ideas and for problem solving, centred on the customer.

2. Which competence does this module give (level)?
Develop new ideas & solutions from a given problem using "design thinking" (level 3: apply).

3. Which step of System Engineering is Ideation / Design Thinking?
Step 5 (Ideation) — after Risk & Change, before Design & Test.

4. The three things design thinking puts in focus?
Customer / benefit, possibility to realize (feasibility), and commercial viability.

5. How should the team be composed?
As heterogeneous (diverse) as possible.

6. What kind of method is design thinking?
An agile method that needs several iterations to reach the goal.

7. Core motto of design thinking?
"Fail fast and cheap!"

8. The two "spaces" in the process?
Problem Space (understand problem & customer) and Solution Space (create & test solutions).

9. The six steps of the process in order?
(1) Understand, (2) Observe, (3) Design Principles, (4) Ideation, (5) Prototyping, (6) Testing.

10. Which steps are the Problem Space?
Understand & Observe (1–2); step 3 bridges into the Solution space.

11. Which steps are the Solution Space?
Ideation, Prototyping, Testing (4–6).

12. Linear or iterative?
Iterative — loop through prototyping & testing several times.

13. (1) Understand — content?
Understand the problem and the target; build the team.

14. In Understand, what must you describe instead of the solution?
The target condition after using the technology — NOT the technical solution.

15. How is the Understand step carried out?
Initial expert circle / brainstorming, updated with each new piece of information.

16. (2) Observe — content?
Understand the customer: their will, wishes & inner needs; collect as much info as possible.

17. What info do you collect about the customer?
The actual problem, what it must do, and personal background (family, hobbies…).

18. How is the Observe step carried out?
Talk to & interview the customer, spend time / observe them, find the relevant peer groups.

19. Why are extreme voices valuable?
They raise important info the average customer wouldn't specifically point out.

20. Two key interview rules (Observe)?
Let the customers talk, not the interviewer; and don't sell your own idea.

21. What does the competency level scale measure?
The persona's (technology) knowledge, 0 (none) to 5 (expert).

22. Competency level 5 (Expert)?
Specialist; handles complex tasks alone; constantly advances the state of the art.

23. (3) Design Principles — content?
Synthesis of steps 1 & 2: find patterns / common sense between customers.

24. How do you capture the customer in Design Principles?
Create a profile using Personas or Point-of-View; visualize the customer's view.

25. What must you update during Design Principles?
Your "(1) Understand" — problem, target, needed functions/features.

26. What is a Persona?
A concept from analytical psychology describing the outer view/role (face) of a person.

27. What is a Buyer Persona?
Persona adapted to economics; a fictional but fitting person describing the target customers.

28. Purpose of a Buyer Persona?
Describe the customer fully so you can take decisions in their name (would he like/use it?).

29. Helpful mindset for writing a persona?
Imagine you write a novel and describe the person (like a good actress).

30. Persona card demographic fields?
Name, Sex, Age, Domicile, Relationship, Kids, Job, Income, Education (fictional but fitting).

31. The four "feeling" blocks on a persona card?
Motivates, Frustrates, Goals, Pain Points / Problems.

32. How many core technologies on the persona card?
Knowledge about 5 relevant core technologies.

33. Who is the example persona Jack Johnsen?
A 54-yr married lifeguard at the Baltic Sea, 3 kids, ~3,150€/month, trained paramedic.

34. (4) Ideation — content?
Create ideas to solve the customer's pain points.

35. The boundary rule in Ideation?
NO boundaries (cost, tech...) — the crazier the better.

36. How is Ideation carried out?
Individual creation → gather & discuss → decide via tools → get customer feedback.

37. Two decision-making tools in Ideation?
Morphological box and QFD (Quality Function Deployment).

38. Criteria used to select ideas?
Practicability, efficiency and economy.

39. (5) Prototyping — content?
Create a tangible, physical prototype that can be experienced.

40. Premise for building a prototype?
Everything is allowed — but the simpler, the better!

41. Key rule when giving a prototype to the customer?
NO verbal explanation; the customer must understand it; it must solve their problem, not realize your idea.

42. (6) Testing — content?
The customer tests the prototype to get feedback; if it fails, change or discard next cycle.

43. How is Testing carried out?
No demonstration; visually observe the customer using it; take several iterations.

44. The famous MRI example?
An MRI was scary for children, so it was redesigned (playful) so kids accept it — solving the real fear.

45. What does the MRI example teach?
Solve the customer's real problem (the child's fear), not only the technical one.

46. Workshop: how is the class organised?
Into 5 teams; each team one set of material, each person a stack of paper templates.

47. Workshop interview rule (Observe)?
Interview min. 2 customers from OTHER groups (not your own); describe the problem only.

48. Workshop Ideation-create rule?
Drawing/painting only — NO writing, NO exchange; min. 5 (better 10) ideas.

49. How many ideas are prototyped?

2 ideas, chosen as a group from customer feedback.

50. Total time of the workshop?

140 min base + 10 min per team = 190 min.

51. What must the final presentation answer?

Problem? Personas? Ideas you went through? Final solution?

52. Summary — design thinking in one line?

A customer-centred method that solves customer needs through fast, iterative cycles of ideas.

EXAM REHEARSAL · WRITE FIRST, THEN REVEAL

3 Practice Questions + model answers

Try to answer on paper before clicking **Reveal**. The model answers show the exact points an examiner rewards.

Q1 **Describe the design thinking process. Name and explain its six steps, and state which belong to the problem space and which to the solution space.**

TYPE: STRUCTURED / THEORY · ~12 MARKS

REVEAL MODEL ANSWER

✓ **MODEL ANSWER**

Design thinking is a **customer-centred**, **agile** method moving between a **Problem Space** and a **Solution Space**, in an **iterative** loop (fail fast and cheap).

- (1) **Understand** (problem space): build the team and understand the problem & the **target condition** (the outcome, not the solution); who is the customer; environment conditions.
- (2) **Observe** (problem space): understand the customer — interview & observe them, collect their wishes/inner needs and background; let the customer talk, don't sell your idea.
- (3) **Design Principles** (the bridge): synthesise steps 1–2, find patterns, build a **persona** / **point-of-view**, and update the "Understand".
- (4) **Ideation** (solution space): create ideas with **no boundaries** (crazier = better), then consolidate, decide (e.g. morphological box, QFD) and get feedback.
- (5) **Prototyping** (solution space): build a simple, tangible prototype; hand it over with **no explanation** — it must solve the customer's problem, not realise your idea.
- (6) **Testing** (solution space): the customer tests it; **observe without demonstrating**, gather feedback, then **iterate** back to prototyping.

Q2

What is a (buyer) persona and why is it used? List the information a persona card contains, and explain the competency-level scale.

TYPE: THEORY / DEFINITIONS · ~10 MARKS

REVEAL MODEL ANSWER

✓ MODEL ANSWER

A **persona** is a concept from **analytical psychology** describing the outer view/role of a person. A **buyer persona** adapts this to economics: a **fictional but fitting person** that best represents the target customers. You describe them as fully as possible (like writing a novel character) so the team can **take decisions in their name** — “would he/she like it, use it, deal with it?”.

A persona card holds: a **photo/drawing**; demographics (**Name, Sex, Age, Domicile, Relationship, Kids, Job, Income, Education**); knowledge of 5 **core technologies**; and four blocks — **Motivates, Frustrates, Goals, Pain Points/Problems**.

The **competency-level scale** (**0–5**) rates the persona's technology knowledge: **0** none · **1** Newbie · **2** basic · **3** advanced · **4** consolidated · **5** expert (advances the state of the art).

Q3 **Maya wants to build a gadget for Jack, a lifeguard who loves surfing. (a) What are the rules for Observe (interview) and for Prototyping/Testing? (b) Why is the MRI example a good design-thinking story? (c) Name two decision-making tools and the three foci of design thinking.**

TYPE: APPLICATION · ~8 MARKS

REVEAL MODEL ANSWER

✓ MODEL ANSWER

(**a**) **Observe:** talk to and observe the customer, find peer groups, value **extreme voices**; crucially **let the customer talk and don't sell your own idea. Prototyping/Testing:** build a simple tangible prototype, hand it over with **no verbal explanation/demonstration**, **observe** the customer using it, gather feedback, and **iterate** — the customer (not you) must understand it, and it must solve their problem.

(**b**) **MRI:** the scanner frightened children, so it was redesigned to be playful/un-scary. It shows design thinking solves the customer's **real** problem (the child's fear) rather than only the technical one — customer-centred and iterative.

(**c**) Decision-making tools: **morphological box** and **QFD**. The three foci: **customer/benefit**, **possibility to realize** (**feasibility**), and **commercial viability**.

LAST 5 MINUTES BEFORE THE EXAM

One-screen cheat-card

6 steps	**Understand·Observe·Design-principles·Ideate·Prototype·Test**	Maya's beach → sea walk
Two spaces	**Problem (1-2) / Solution (4-6)**	dry sand vs water; persona = bridge
3 in focus	**Customer · Feasibility · Viability**	surfboard on 3 rocks
Mindset	**agile · several iterations · fail fast & cheap**	snap cardboard, grab another
Team	**as heterogeneous as possible**	mixed volleyball team
(1) Understand	**problem + target (NOT the solution!)**	jigsaw “?”
(2) Observe	**learn from customer; let them talk**	telescope + zipped mouth
Extreme voices	**reveal more than the average customer**	the loud seagull guy
Competency 0-5	**0 none → 5 expert**	lifeguard tower
(3) Persona	**synthesis; fictional customer; update Understand**	cardboard Fake-Jack
Persona vs Buyer	**role/mask vs economic customer model**	Greek theatre mask
(4) Ideation	**no limits, crazier = better; pick with tool**	lightbulb storm
Decision tools	**morphological box · QFD**	slot machine of boxes
(5) Prototype	**tangible, simple, NO explanation**	duct-tape + silence
(6) Testing	**hand over, observe, iterate**	palm-tree spy
Example	**MRI made un-scary for kids**	scary tube → pirate sub